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Smart Cruise E-Bike

(Final Project Report)



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1. Introduction

With the growing demand for sustainable energy solutions and environmentally friendly transportation, electric bicycles have emerged as an effective alternative to conventional vehicles. This project focuses on converting a regular bicycle into a fully functional electric bicycle using an efficient and cost-effective system.

In addition to providing electric assistance for transportation, the project explores a secondary system that allows the bicycle to generate electricity when lifted on a stand, which can be used to charge mobile devices.

The project aims to combine mechanical and electrical engineering principles to develop a practical and sustainable solution. This final report presents the complete development of the project, including system selection, methodology, implementation, challenges, and future improvements.

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2. Project Description

This project involves converting a regular bicycle into an electric bicycle. By using a hub motor system, the bike provides electrical help to the rider. The system uses a hub motor, battery, controller, throttle, and a Smart PID Cruise Control system that all work together to make the ride smooth, precise, and easy.

The project was completed in a few steps: researching the best systems, buying the parts, installing everything, and testing the bike. Great care was taken to pick parts that are safe, work well, and do not cost too much money.

The Smart PID Cruise Control system is built around five components: an ESP32 microcontroller acting as the PID controller brain, an X9C103S digital potentiometer controlling the throttle signal programmatically, a Buck Converter (XL7015) stepping down the 48V battery to a stable 5V to power the ESP32 and relay, a Digital Speedometer feeding real-time speed back to the ESP32 as the measured process variable, and a Relay Module switching the signal source between the manual throttle and the X9C103S depending on whether cruise control is active or not.

The design is built to allow for several future updates. These include a system to generate electricity for charging phones while pedaling on a stand, a screen to show the current speed and battery charge, and further fine-tuning of the PID gains for even more precise speed regulation across different terrain and load conditions.

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3. Project Objectives

- Convert a conventional bicycle into an electric bicycle using an efficient and reliable hub motor system.
- Study and compare different conversion systems and select the most suitable option based on performance, cost, and availability.
- Implement a Smart PID Cruise Control system that allows the rider to maintain a precise set speed without holding the throttle, using a closed-loop feedback control algorithm running on an ESP32 microcontroller.
- Integrate a Relay Module to seamlessly switch control between the manual throttle and the PID-controlled X9C103S digital potentiometer depending on the cruise control state.
- Apply mechanical and electrical engineering principles in a practical and sustainable project.
- Evaluate the performance, efficiency, and limitations of the implemented system.

4. System Description

4.1) Hub Motor System (System 1) – 450 W

The core of the electric bicycle system is a 450 W Brushless DC Hub Motor integrated directly into the rear wheel. Electrical energy stored in the lithium-ion battery is supplied through the controller to the hub motor, which assists the rider during pedaling.

4.1.1) Hub Motor

The hub motor receives its power from the driver (controller) through two groups of wires:

- **Phase Wires (3 wires):** Carry the three-phase AC power from the controller to the motor windings, responsible for producing torque and driving the wheel.
- **Hall Sensor Wires (6 wires):** Provide real-time rotor position feedback to the controller, enabling precise commutation and smooth motor operation.



Figure (1): Hub Motor

Motor Type	Brushless DC Hub Motor
Rated Power	450 W
Rated Voltage	48 V
Rated Speed	350 – 450 RPM
Efficiency	Approximately 80 – 85%

4.1.2) Driver (Motor Controller)

The driver (controller) is the central electronic unit that connects and manages all system components. It receives power from the battery and distributes regulated power to the motor. Its wire groups include:

- **Power Wires:** Connect to the 48V battery (positive and negative).
- **Motor phase Wires:** Three-phase output going to the hub motor.
- **Hall sensor Wires:** Feedback signals from the motor's Hall sensors.
- **Throttle Wires:** Three wires (GND, VCC, Signal). GND and VCC connect to the throttle grip as usual. The Signal wire connects to the COM terminal of the Relay Module, which switches between the manual throttle signal and the X9C103S wiper output depending on the cruise control state.
- **3-Speed Wires:** Connect to the speed selection switch on the throttle assembly.
- **LCD/Instrumental Cable:** The speed signal output from the driver, connected to the Digital Speedometer signal wire which feeds real-time speed back to the ESP32 as the PID feedback variable.
- **Low Brake & High Brake Wires:** Cut-off signals that instantly disable the motor when brakes are applied and simultaneously trigger the ESP32 to reset the PID loop and return the relay to its default NC state.
- **Anti-Theft Wires:** Security function to disable the motor when needed.
- **Learning Wires:** Used to calibrate the controller with the motor.

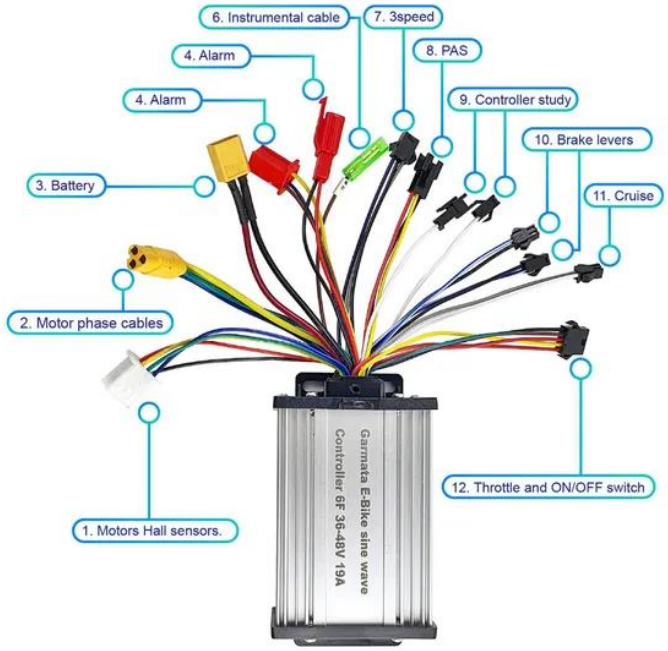


Figure (2): Driver (Motor Controller)

Controller Type	Brushless DC Motor Controller
Operating Voltage	48 V
Functions	Speed control, overcurrent protection, voltage regulation, cruise control

4.1.3) Rechargeable Battery

The battery provides the electrical energy required to drive the hub motor. For safety, a large disconnect switch (similar to air-conditioner switches) is installed on the positive terminal to handle the high current safely. The battery is charged via a dedicated charger.



Figure (3): Rechargeable Battery

Battery Type	Lithium-Ion (Li-ion)
Rated Voltage	48 V
Capacity	10 Ah
Charging Time	Approximately 5 hours (full charge)
Safety Switch	Large disconnect switch on positive terminal

4.1.4) Twisting Throttle

The throttle is mounted on the handlebar and allows the rider to control the motor speed. It contains three main wires: Ground (GND), Supply Voltage (VCC), and Signal. The GND and VCC connect directly to their corresponding wires in the driver as usual. The Signal wire connects to the NC terminal of the Relay Module, so when cruise control is inactive the manual throttle signal passes through the relay directly to the driver's signal input, giving the rider full manual speed control.

In addition, it houses the 3-speed switch connected to its corresponding wire group in the driver as usual, and the cruise control button which is now connected directly to a GPIO input pin on the ESP32 instead of the driver's cruise wires, so that pressing it triggers the ESP32 to capture the current speed as the set-point and activate the PID loop.



Figure (4): Twisting Throttle

4.2) Smart PID Cruise Control System

One of the key additions to this project is the implementation of a Smart PID Cruise Control system. Unlike a simple speed-lock mechanism, this system uses a closed-loop PID (Proportional-Integral-Derivative) controller running on an ESP32 microcontroller to actively regulate the motor speed. This enables the rider to maintain a precise, constant speed without continuously holding the throttle, even across varying terrain and load conditions.

4.2.1) System Overview

The system is composed of four main hardware components working together to form a closed-loop speed control loop:

1. **ESP32 Microcontroller:** Acts as the PID controller brain.
2. **X9C103S Digital Potentiometer:** Replaces the manual throttle signal, allowing the ESP32 to programmatically control motor speed.
3. **Buck Converter (XL7015):** Steps down the 48V battery voltage to a stable 5V to safely power the ESP32.
4. **Relay Module (5V Single Channel):** Acts as an electronically controlled switch between the manual throttle signal and the X9C103S wiper output. When cruise is inactive it stays in its default NC state passing the manual throttle signal directly to the driver, and when cruise is activated the ESP32 flips it to NO disconnecting the manual throttle and connecting the X9C103S wiper to the driver's signal input instead.
5. **Digital Speedometer:** Reads the speed signal from the driver's LCD display output wire and feeds it back to the ESP32 as the measured process variable.

4.2.2) ESP-32 Microcontroller

The ESP32 is a powerful, low-cost microcontroller with built-in Wi-Fi and Bluetooth. In this project it serves as the PID controller, reading the current speed from the digital speedometer and adjusting the X9C103S digital potentiometer output to regulate motor speed. The PID algorithm continuously computes the error between the set-point speed and the actual speed, then corrects accordingly

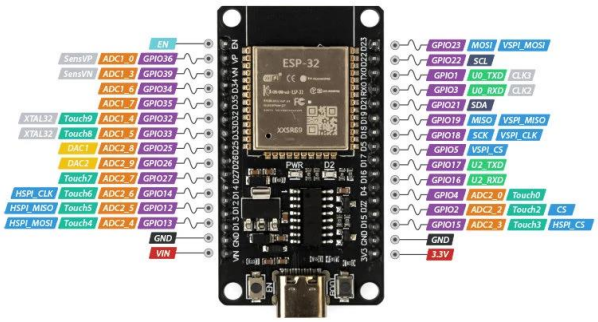


Figure (5): ESP-32 Microcontroller

Microcontroller	ESP-32
Operating Voltage	3.3 V / 5 V tolerant inputs
Supply Voltage (in project)	5 V (from Buck Converter)
Role	PID Controller; reads speed, outputs correction via digital potentiometer

4.2.3) Digital Potentiometer (X9C103S)

The X9C103S is a digitally controlled potentiometer with 100 resistance steps (0–10 kΩ). The ESP32 controls its wiper position via SPI-like signals (INC, U/D, CS), which directly adjusts the throttle signal voltage sent to the motor controller. This allows the PID algorithm to increase or decrease motor power without any mechanical intervention.



Figure (6): Digital Potentiometer (X9C103S)

Component	X9C103S Digital Potentiometer
Resistance Range	0 – 10 kΩ (100 steps)
Control Interface	INC, U/D, CS (SPI-like)
Role	Replaces manual throttle; controlled by ESP32 PID output

4.2.4) Buck Converter (XL7015)

The XL7015 buck converter steps the 48V battery supply voltage down to a regulated 5V DC output. This is essential for safely powering the ESP32, which would be permanently damaged by 48V. The converter provides a stable and efficient supply regardless of battery discharge level.



Figure (7): Buck Converter (XL7015)

Component	Buck Converter (XL7015)
Input Voltage	48 V (from main battery)
Output Voltage	5 V (regulated)
Role	Voltage step-down to safely power the ESP32 microcontroller

4.2.5) Relay Module (5V Single Channel)

A single-channel relay module is used as an electronically controlled switch to alternate the throttle signal source between the manual throttle and the X9C103S digital potentiometer. When the cruise control is inactive, the relay stays in its default state (NC – Normally Closed), connecting the manual throttle signal wire directly to the driver's signal input, allowing the rider full manual speed control. When the rider activates cruise control, the ESP32 sends a HIGH signal to the relay coil, switching it to (NO – Normally Open), which disconnects the manual throttle signal and connects the VW (wiper) output of the X9C103S to the driver's signal input instead, handing full control to the PID algorithm.



Figure (8): Relay Module (5V Single Channel)

Operating Voltage	5V (powered from Buck Converter output)
Control Pin	Connected to ESP32 GPIO output pin
NC Terminal	Connected to Signal wire of manual throttle
NO Terminal	Connected to VW (wiper) of X9C103S
COM Terminal	Connected to Signal wire input of the driver
Default State (Cruise OFF)	NC closed — manual throttle in control
Activated State (Cruise ON)	NO closed — X9C103S / PID in control

4.2.6) Digital Speedometer

The digital speedometer is wired directly to the motor controller's LCD display output signal wire (labelled as LCD/Instrumental cable on the controller). Its VCC and GND are connected to the main battery, while its signal wire taps into the speed signal coming from the driver. This signal is fed into the ESP32 as the feedback variable (measured speed) for the PID loop.



Figure (9): Digital Speedometer

Component	Digital Speedometer
VCC / GND	3.3 V / 5 V tolerant inputs
Signal Wire	Connected to LCD/Instrumental signal output of the driver
Role	Provides real-time speed feedback to ESP32 for PID closed-loop control

4.2.7) How it Works

The cruise control function is built around the ESP32 PID controller. The operation sequence is as follows:

1. The rider accelerates to the desired speed using the manual throttle, which is connected through the relay's NC terminal to the driver's signal input.
2. The rider presses the cruise control button on the throttle assembly, which sends a signal to the ESP32 GPIO input pin.
3. The ESP32 captures the current speed reading from the digital speedometer as the set-point, flips the relay from NC to NO, disconnecting the manual throttle signal and connecting the X9C103S wiper output to the driver's signal input instead.
4. The digital speedometer continuously measures the actual speed via the driver's LCD/Instrumental signal wire and feeds it back to the ESP32 as the PID process variable.
5. The ESP32 PID controller calculates the error (set-point minus actual speed) and sends INC and U/D signals to the X9C103S to move its wiper up or down, increasing or decreasing the voltage on the driver's signal input and therefore the motor power accordingly.
6. The wheel continues running at the set speed automatically, with the PID loop continuously correcting for terrain changes, slopes, or load variation.
7. Pressing the brake lever triggers the brake cut-off sensors which instantly disable the motor, and simultaneously signal the ESP32 to reset the PID loop, return the relay to its default NC state, and restore full manual throttle control.

4.2.8) Wiring

- **Buck Converter Input:** Connected to the 48V battery positive and negative terminals.
- **Buck Converter Output (5V):** Connected to the VCC and GND pins of the ESP32 and the VCC and GND of the Relay Module.
- **Digital Speedometer VCC/GND:** Connected to the 48V battery.
- **Digital Speedometer Signal Wire:** Connected to the LCD/Instrumental cable output of the motor controller and to an ESP32 digital input pin as the PID speed feedback variable.
- **X9C103S (VCC, GND):** Connected to the 5V output of the Buck Converter.
- **X9C103S (INC, U/D, CS Pins):** Connected to ESP32 GPIO output pins.
- **X9C103S VH:** Connected to 5V from the Buck Converter.
- **X9C103S VL:** Connected to GND.
- **X9C103S VW (Wiper):** Connected to the NO terminal of the Relay Module.
- **Relay Module COM Terminal:** Connected to the Signal wire input of the driver.
- **Relay Module NC Terminal:** Connected to the Signal wire of the manual throttle.
- **Relay Module NO Terminal:** Connected to the VW (wiper) of the X9C103S.
- **Relay Module Control Pin:** Connected to an ESP32 GPIO output pin.
- **Cruise Control Button:** One wire to an ESP32 GPIO input pin, the other wire to GND.

4.2.9) Safety

- The brake cut-off sensors (Low Brake and High Brake wires) are critical safety features that override the PID cruise control immediately upon braking. When a brake is applied, the cut-off signal disables the motor driver output instantly, preventing the motor from fighting against the brakes and ensuring rider safety regardless of the PID controller state.

4.3) Generation System

The system also integrates a secondary generation circuit that converts mechanical energy from the rear wheel into usable electrical power for charging a mobile device via USB, independent of the main 48V drive battery.

4.3.1) System Overview

The generation system consists of four components forming a complete AC-to-regulated-DC power path:

- **Bicycle Dynamo (Generator):** Converts rotational motion from wheel contact into raw AC voltage.
- **Bridge Rectifier:** Converts the dynamo's AC output into unregulated DC.
- **Smoothing Capacitor:** Filters ripple from the rectified output before regulation.
- **Buck Converter (LM2596):** Steps down the variable rectified voltage to a stable 5V DC output.
- **USB-C Output Socket:** Delivers regulated 5V to charge a mobile device.

4.3.2) Bicycle Dynamo

A friction-type bicycle dynamo generates AC voltage through contact with the tire sidewall as the wheel rotates. Output voltage scales with riding speed.



Figure (10): Bicycle Dynamo

Model	Wai Danie
Rated Output	12V AC, 6W
Output Type	Alternating Current (AC)
Water Proofing	Yes
Body Material	Aluminium Alloy
Role	Converts wheel rotation into raw AC voltage

4.3.3) Bridge Rectifier

A full-wave bridge rectifier converts the dynamo's AC output into pulsating DC, since the buck converter downstream requires a DC input.



Figure (11): Bridge Rectifier

Type	KBP206
Forward Current	2 A
Reverse Voltage rating	600 V
Role	AC to DC conversion

4.3.4) Smoothing Capacitor

An electrolytic capacitor is placed across the rectifier output to reduce voltage ripple before the signal reaches the buck converter, ensuring a more stable input for regulation.



Figure (12): Smoothing Capacitor

Capacitance	1000 µF
Rated Voltage	25 V
Role	Ripple filtering / voltage smoothing

4.3.5) Buck Converter

The LM2596 buck converter regulates the variable rectified input voltage down to a fixed, stable 5V output suitable for USB charging, regardless of fluctuations in riding speed.



Figure (13): Buck Converter

Input Voltage Range	2.5 - 40 V
Output Voltage	5 V (fixed, set via onboard potentiometer)
Output Display	Integrated LED voltmeter
Role	Voltage regulation for safe USB charging

4.3.6) USB-C Output

A waterproof USB Type-C female socket provides the charging interface. Only the power lines (VBUS and GND) are wired, as the system delivers power only, with no data connection required.



Figure (14): Bicycle Dynamo

Type	USB Type-C Female, 2-pin (power only)
Rated Output	5V DC
Protection	Waterproof housing
Role	Mobile device charging interface

4.3.7) Wiring

The dynamo's two AC output leads connect to the rectifier's AC input terminals. The rectifier's DC output (+/−) connects to the smoothing capacitor and the buck converter's input terminals (IN+/IN−). The buck converter's output terminals (OUT+/OUT−) connect to the USB-C socket's power wires, completing the charging path.

5. Methodology

The project methodology was structured to guide the work from research and system evaluation to practical implementation.

5.1) System Research and Selection

- Conducted detailed research on potential conversion systems for the bicycle.
- Compared systems based on performance, installation complexity, cost, and component availability.
- Selected the 450 W Hub Motor system for the e-bike conversion due to its efficiency and practicality.
- Designed the Smart PID Cruise Control system architecture using ESP32, X9C103S digital potentiometer, Buck Converter (XL7015), Digital Speedometer, and a Relay Module.

5.2) Component Procurement

- Prepared a complete list of required components for both systems.
- Sourced components from local and online suppliers at reasonable cost.

5.3) System Installation

- Mounted the hub motor on the rear wheel and installed the battery and controller.
- Connected the full wiring harness including throttle, Hall sensors, and brake cut-offs.
- Installed the Buck Converter to step down 48V to 5V for powering the ESP32 and Relay Module.
- Connected the Digital Speedometer to the driver's LCD/Instrumental signal output for speed feedback.
- Wired the X9C103S digital potentiometer between the ESP32 and the Relay Module NO terminal.
- Wired the Relay Module COM to the driver's signal input, NC to the manual throttle signal, and control pin to the ESP32 GPIO.
- Connected the cruise control button to the ESP32 GPIO input pin.

5.4) Testing and Initial Operation

- Performed initial testing of the hub motor system under load — confirmed correct operation.
- Verified manual throttle control works normally through the relay NC state before cruise activation.
- Tested the Smart PID Cruise Control by reaching a set speed, pressing the cruise button, and confirming the ESP32 flips the relay and maintains the set speed automatically through the X9C103S.
- Verified that brake cut-off sensors immediately reset the PID loop and return the relay to NC state restoring manual throttle control.
- Adjusted PID gains (Kp, Ki, Kd), mounts, wiring, and connections to ensure safety and reliability.

6. Progress Achieved

6.1) Hub Motor System

- 450 W hub motor successfully mounted on the rear wheel.
- Battery and controller installed and wired correctly.
- Electric assistance confirmed — smooth acceleration up to 25–30 km/h.
- Estimated range of 40–50 km per charge validated.

6.2) Smart PID Cruise Control System

- Buck Converter successfully stepping 48V down to stable 5V for powering the ESP32 and Relay Module.
- Digital Speedometer wired to the driver's LCD/Instrumental signal output and confirmed sending real-time speed feedback to the ESP32.
- Relay Module wired and confirmed switching correctly between manual throttle (NC) and X9C103S wiper (NO) upon cruise activation and deactivation.
- X9C103S digital potentiometer confirmed responding to ESP32 INC and U/D signals and producing correct voltage on the driver's signal input.
- Smart PID Cruise Control fully tested — rider accelerates to desired speed using manual throttle, presses cruise button, relay switches to NO, ESP32 locks the set-point and maintains the speed automatically through the PID loop and X9C103S.
- Brake cut-off sensors confirmed to instantly disable the motor, reset the PID loop, and return the relay to NC state restoring full manual throttle control.

6.3) Component Integration

- All components for both systems procured and integrated without major issues.
- Safety measures including proper wiring, secure mounting, and brake cut-offs fully implemented.



Figure (15): Progress Acheived

7. Conclusion

This project successfully demonstrates the conversion of a conventional bicycle into a fully functional electric bicycle using a 450 W hub motor system, complete with a Smart PID Cruise Control feature. The hub motor system provides efficient electric assistance, while the Smart PID Cruise Control — built around an ESP32, X9C103S digital potentiometer, XL7015 Buck Converter, Digital Speedometer, and a Relay Module — actively maintains a set speed through closed-loop feedback without requiring continuous throttle input.

The project highlights the feasibility of combining mechanical and electrical engineering principles to achieve a practical, low-cost, and environmentally friendly solution, establishing a strong foundation for further innovation in sustainable mobility.

Future development will focus on fine-tuning PID gains, adding a digital display, exploring ESP32 Bluetooth integration for remote parameter tuning, and implementing a pedal-powered electricity generation system for mobile device charging — further enhancing the project's utility as an intelligent and energy-efficient transportation platform.



Figure (16): Smart Cruise E-Bike